



MFTrans: A multi-feature transformer network for protein secondary structure prediction

Yifu Chen^a, Guanxing Chen^a, Calvin Yu-Chian Chen^{a,b,c,d,e,*}

^a Artificial Intelligence Medical Research Center, School of Intelligent Systems Engineering, Shenzhen Campus of Sun Yat-sen University, Shenzhen, Guangdong 518107, China

^b AI for Science (AI4S)-Preferred Program, Peking University Shenzhen Graduate School, Shenzhen, Guangdong 518055, China

^c School of Electronic and Computer Engineering, Peking University Shenzhen Graduate School, Shenzhen, Guangdong 518055, China

^d Department of Medical Research, China Medical University Hospital, Taichung 40447, Taiwan

^e Department of Bioinformatics and Medical Engineering, Asia University, Taichung 41354, Taiwan

ARTICLE INFO

Keywords:

Protein secondary structure prediction

Multiple sequence alignment

Feature fusion

Deep learning

ABSTRACT

In the rapidly evolving field of computational biology, accurate prediction of protein secondary structures is crucial for understanding protein functions, facilitating drug discovery, and advancing disease diagnostics. In this paper, we propose MFTrans, a deep learning-based multi-feature fusion network aimed at enhancing the precision and efficiency of Protein Secondary Structure Prediction (PSSP). **This model employs a Multiple Sequence Alignment (MSA) Transformer in combination with a multi-view deep learning architecture to effectively capture both global and local features of protein sequences.** MFTrans integrates diverse features generated by protein sequences, including MSA, sequence information, evolutionary information, and hidden state information, using a multi-feature fusion strategy. The MSA Transformer is utilized to interleave row and column attention across the input MSA, while a Transformer encoder and decoder are introduced to enhance the extracted high-level features. A hybrid network architecture, combining a convolutional neural network with a bidirectional Gated Recurrent Unit (BiGRU) network, is used to further extract high-level features after feature fusion. In independent tests, our experimental results show that MFTrans has superior generalization ability, outperforming other state-of-the-art PSSP models by 3 % on average on public benchmarks including **CASP12, CASP13, CASP14, TEST2016, TEST2018, and CB513.** Case studies further highlight its advanced performance in predicting mutation sites. MFTrans contributes significantly to the protein science field, opening new avenues for drug discovery, disease diagnosis, and protein.

1. Introduction

Proteins play a crucial role in the vital processes of all living entities, with their functionality intimately linked to their structural formation. This underscores the importance of exploring protein sequences, their structural configurations, and their spatial organization to unravel their biological roles [1]; [2]. The elucidation of protein functionalities and the interactions among various proteins hinge on a comprehensive understanding of their specific structural attributes, highlighting the critical need for protein structure research. Protein structures are categorized into four levels: primary, secondary, tertiary, and quaternary, with the tertiary structure being biologically active. Currently, X-ray crystallography and nuclear magnetic resonance are the predominant techniques for investigating the tertiary protein structure [3].

However, these methodologies for structural elucidation face drawbacks such as significant expense and extended durations. With the advancement of sequencing technology and related research, the number of protein primary structure sequences (i.e., the amino acid sequence of protein composition) has exponentially increased, with many of these sequences stored in large biological databases. Generally, there are two approaches to predict the tertiary structure of proteins: inputting secondary structure for prediction and directly inputting primary structure. Yet, due to the complex arrangement and diverse structure of amino acids in the tertiary structure, predicting it directly from the primary structure is extremely challenging, often necessitating the use of secondary structure prediction results. Protein secondary structure, describing the local spatial structure of a protein, serves as an intermediary connecting primary and tertiary structures, thereby aiding in

* Corresponding author at: AI for Science (AI4S)-Preferred Program, Peking University Shenzhen Graduate School, Shenzhen, Guangdong 518055, China.

E-mail address: cy@pku.edu.cn (C.Y.-C. Chen).

<https://doi.org/10.1016/j.ijbiomac.2024.131311>

Received 7 May 2023; Received in revised form 21 March 2024; Accepted 30 March 2024

Available online 9 April 2024

0141-8130/© 2024 Elsevier B.V. All rights reserved.

predicting protein tertiary structure [4]. Thus, improving the accuracy of protein secondary structure prediction is a critical task in structural biology.

Protein secondary structure prediction (PSSP) is tasked with categorizing each amino acid in a protein sequence by assigning secondary structure tags to describe local structures. These secondary structures, characterized by patterns of hydrogen bonds along the protein's main chain, lead to the formation of local configurations predominantly as **alpha-helices (H), beta-strands (E), and coils (C)** [5]. Initial approaches to PSSP relied on statistical analyses of amino acid residues' propensity for forming specific structures, evolving into more sophisticated statistical models over time. However, these early models were not sufficiently effective to fulfill practical requirements [6]. The advent and progression of artificial intelligence technologies have heralded a new era for PSSP, with the introduction of advanced machine learning (ML) and deep learning (DL) techniques marking a notable leap forward in the field [7–9]. In this innovative landscape, numerous DL-based strategies have been introduced for predicting protein secondary structure. One notable method, **SPOT-1D** [10], integrates diverse DL architectures, primarily leveraging a synergy of bidirectional recurrent neural networks and bidirectional gated recurrent unit (BiGRU) networks, augmented by a residual convolutional neural network (CNN). This approach not only focuses on the local interactions among amino acids but also extends its analysis to capture relationships between more distantly positioned amino acids, thereby achieving a marked enhancement in prediction accuracy.

DeepCNF [11] utilizes a synergistic approach that merges deep CNNs with **conditional random fields (CRFs)** to determine the global optimal solution by assessing the correlations between amino acid residues and their labels. This method has demonstrated a high level of precision, achieving Q3 accuracies of 84.4 % and 84.7 % on the CASP10 and CASP11 datasets, respectively. **MUFOLD-SS** [12], **leveraging the Deep3I network architecture**, incorporates two nested initial modules alongside a convolutional and a fully connected (FC) layer. This design enables effective convolution operations to manage both local and global interactions among amino acid residues, thereby enhancing prediction accuracy. **PSRSM** [13] employs a strategy rooted in data segmentation and semi-random subspace techniques, categorizing the training dataset into various subsets based on protein length. A supported vector machine (SVM) is utilized as the primary classifier, with the ultimate classifier emerging from an aggregation of votes. **GTPCs** [14] **introduces a novel methodology centered around group template pattern classifiers capable of identifying long-range protein interactions. Feature extraction is performed using wavelet transformation on protein sequences, which is then integrated with SVM classification to deliver the final predictive outcomes.** Senior et al. [15] have developed a neural network model adept at analyzing intervals between amino acid residues, thus predicting more structural details and achieving improved accuracy for sequences with fewer homologues [15]. Furthermore, Hu et al. [16] have introduced an ensemble model based on a BiGRU neural network framework. This ensemble consists of five distinct segments, each processing different input features. Each segment is associated with a specific loss function, and their collective outputs are combined to derive the final classification result.

These methods heavily rely on sequence information for predicting protein secondary structure. Integrating additional secondary structure information is expected to enhance prediction accuracy. AlphaFold2 [17] has achieved unparalleled accuracy in predicting protein structures directly from their amino acid sequences, primarily due to its adept extraction of coevolutionary signals through MSA. The use of coevolutionary data for inferring contacts and modeling structures has been a common approach in research. However, earlier efforts were often hampered by the weak or overly complex coevolutionary signals within the MSAs. Rao et al. [18] introduced a novel approach with the MSA transformer, an unsupervised protein language model that processes the MSA of a query sequence instead of relying on a singular sequence. This

model employs both row and column attention mechanisms on the input MSAs and utilizes objectives akin to masked language modeling. By using the attention mechanism, it becomes possible to accurately isolate genuine residue signals from the MSAs, emphasizing the importance of incorporating MSA characteristics into the PSSP task to improve model predictive performance.

In this paper, we introduce MFTrans, as illustrated in Fig. 1. MFTrans is a deep learning-based PSSP framework that combines a multi-view DL architecture with the MSA Transformer, aiming to fully assimilate both global and local features inherent in protein sequences. Specifically, this paper has the following critical contributions:

- We propose MFTrans, a novel approach for PSSP, combining four distinct feature representation strategies to dissect various sequential data insights.
- We utilize an MSA Transformer, incorporating row and column attention mechanisms within the input MSA, coupled with a synergistic framework that merges CNN and BiGRU networks. This architecture is designed to effectively blend features and excavate deep-latent attributes.
- Through rigorous benchmarking against standard datasets, MFTrans is demonstrated to outperform contemporary leading-edge methodologies in independent assessments. It consistently surpasses other models by an average margin of 3 % across public benchmarks, including CASP12, CASP13, CASP14, TEST2016, TEST2018, and CB513, showcasing its superior performance.

The remainder of this article is organized as follows: Section 2 presents a comparison and discussion of related work. Section 3 details the proposed framework and method. Section 4 describes the datasets and implementation details of our model. Section 5 primarily contains experimental results and corresponding analysis discussions. Finally, we summarize our work in Section 6.

2. Related work

2.1. Multiple sequence alignment (MSA)

MSA involves aligning multiple homologous protein sequences for a target protein [19]. Given a protein database and sequence, MSA identifies alignments within the database through pairwise comparisons [20], employing probabilistic models like hidden Markov models [21–23], or both approaches combined [24]. It is crucial for modeling sequence relationships in computational biology. Recent studies have trained deep variational autoencoders on MSA to predict functions [25] and demonstrated that autoregressive models can learn protein function from sequences without alignments [26]. **Various protein language models have been proposed to extract evolutionary information from MSA [27–29]. The MSA Transformer model, proposed by Rao et al. [18], is an unsupervised protein language model that processes MSAs of query sequences. This model, utilizing row and column attention and a masked language modeling objective, has shown accuracy in predicting long-range contacts between residues and other protein properties, such as secondary structure prediction and mutational effects.**

2.2. Position-specific scoring matrix (PSSM)

With the rapid increase in proteins of unknown structures and functions, comparing protein sequences of unknown structure with those of known structure has become a key research approach [30]. Therefore, the similarity of protein sequences has become an important starting point for researchers to study protein structure and function. From this perspective, there is still a lot of useful information to be extracted by comparing protein sequences of unknown structure with known protein sequences. The PSI-BLAST tool [24] is often used to compare the similarity of protein sequences in databases to obtain the

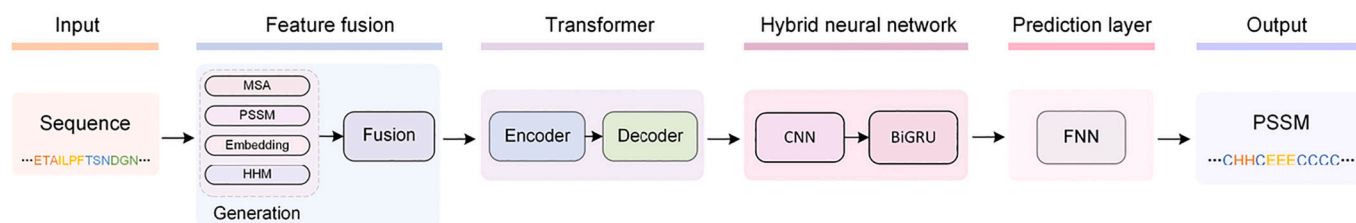


Fig. 1. The flowchart of MFTrans. MFTrans processes protein sequences to produce four distinct types of features: MSA feature, PSSM feature, HMM feature, and sequence embedding feature. Following the fusion of these multiple features, the combined output is then fed into a seamless end-to-end network architecture, culminating in the production of the final PSSP outcome.

resulting PSSM profile. The commonly used research method is to construct PSSM as the input of the classifier. The proposal of PSSM has effectively improved the classification accuracy of 3-state secondary structure classification. Online protein secondary structure classification servers, including PHD [31], PSIPRED [32], Jpred [33], and RaptorX [34], utilize PSSM and MSA profiles as inputs, with the PHD method serving as a benchmark for PSSP and achieving a three-state classification accuracy (Q3) of over 80 %.

2.3. Hidden Markov model (HMM)

A hidden Markov model is a dynamic statistical model that can be described by a finite state machine [35]. HMMs are particularly suitable for processing one-dimensional sequences. They can be trained to obtain probabilistic transition matrices for each state, thus describing different

protein families. The principle of HMM analysis of protein sequences is to analyze the probability of generating different sequences. HMMs can calculate the probability that the model produces different sequences. For sequences that are consistent with the model, the probability of generating sequences according to this model will be higher. If the sequence is not consistent with the model, the probability of generating the sequence according to this model will be lower. The probability output of the HMM can be used to predict the secondary structure of the sequence. In PSSP, HMMs learn structures such as alpha-helix (H), beta-strand (E), and coil (C), and apply these HMMs to new protein sequences with unknown secondary structures. Liu et al. [36] proposed a protein binding site prediction method that combined sequential contour trends and hidden Markov support vector machine (HM-SVM) prediction results. The HMMER 3.3.2 tool [37] is commonly used to generate HMM profiles from protein sequences.

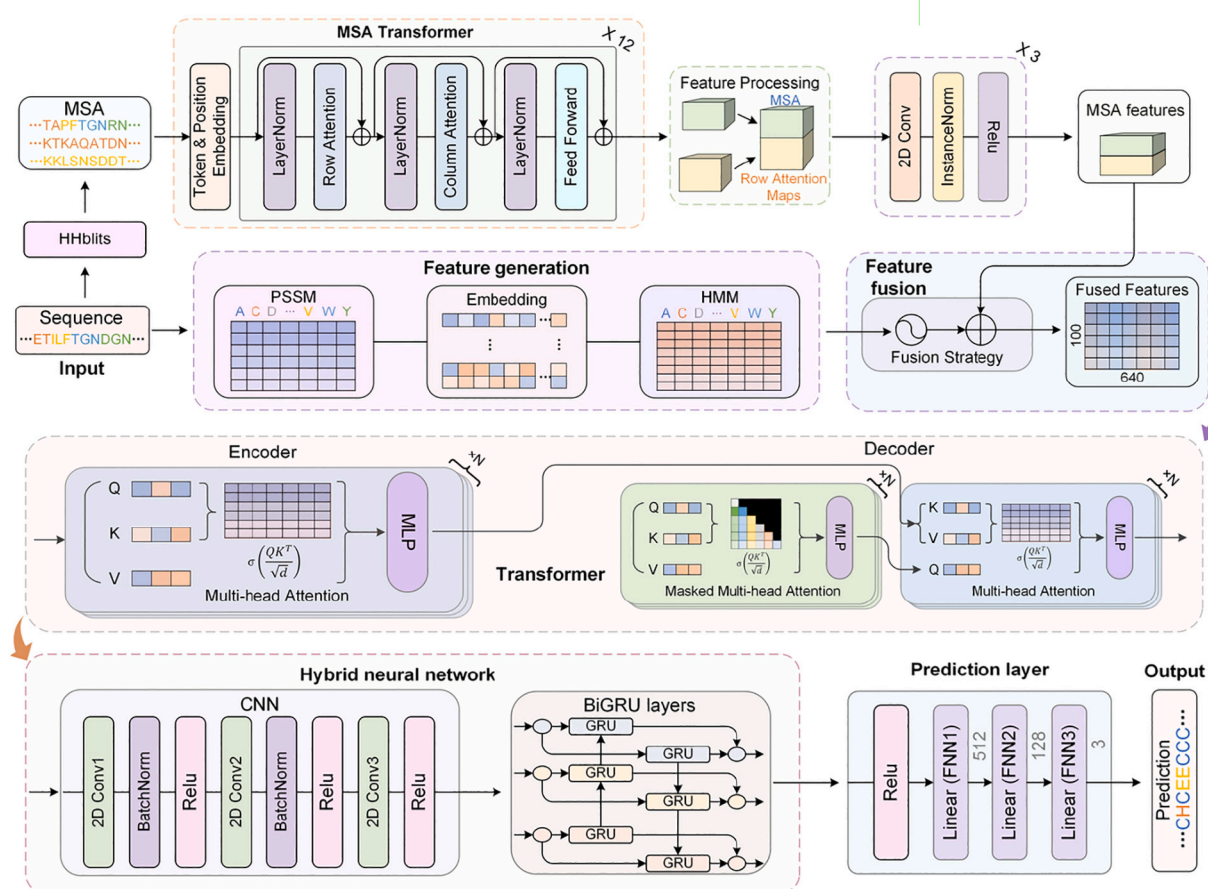


Fig. 2. The MFTrans framework is structured as follows: (1) It begins by accepting a protein sequence, from which it derives four distinct feature types. MFTrans then amalgamates these into a singular fused feature, enabling the exploration of various sequences information. (2) Employing a composite architecture that combines transformer and BiGRU networks, it processes the fused feature to unveil high-latent features subsequent to fusion. (3) The refined features are then inputted into a pre-trained prediction module, which is responsible for generating the ultimate PSSP output.

3. Method

3.1. Overview of the method

Fig. 2 depicts the structure of our MFTrans model, encompassing four primary components: (1) the multi-feature fusion module, (2) the feature extraction module, (3) the feature enhancement module, and (4) the prediction module. Utilizing the MSA Transformer [18], a pre-trained model educated on 26 million MSAs, we derive features from MSA profiles that are assembled from protein sequences. The multi-feature fusion module amalgamates four types of feature metrics: MSA feature, PSSM feature, HMM feature, and sequence embedding feature, employing a fusion technique based on cosine similarity [38]. To extract and enhance features, we employ a composite neural network framework combining CNN and BiGRU to capture both local and global feature dimensions from the merged features. An implementation of a Transformer encoder [39] processes the intermediate features prior to their allocation to the prediction module, which subsequently predicts the secondary structure for each amino acid position. The functionalities of each module are elaborated upon further in the sections that follow.

3.2. Multi-feature fusion

In this section, we delineate the methods employed for preprocessing protein sequences, facilitating the creation of diverse feature representations suitable for training in machine learning frameworks. We present four distinct categories of features: MSA feature, PSSM feature, HMM feature, and Sequence Embedding Feature. Furthermore, we elucidate the embedding techniques utilized to derive each feature type. To amalgamate these features into a cohesive feature space, we propose a fusion strategy based on Cosine Similarity [38].

3.2.1. MSA feature generation

To analyze a specific protein sequence, we employ HHblits version 3.3.0 in conjunction with the unclust30_2018_08 database [40] for generating MSA profiles for the sequences under investigation. Subsequent to this, we harness evolutionary insights through the MSA Transformer. The input MSA is conceptualized as a character matrix with dimensions $r \times c$, where r signifies the number of sequences (rows) and c denotes the sequence length (columns) within the MSA. Upon processing with the MSA Transformer, the input MSA is transformed into two distinct feature types: (1) Initially, it is represented as a tensor of dimensions $r \times c \times 768$. This tensor undergoes dimensional reduction through three linear layers, eventually compacting to a 128-dimensional feature. Following a series of horizontal and vertical expansions and stacking processes, this dimensionally reduced feature forms a query sequence feature, delineated as a tensor with dimensions $c \times c \times 256$, and is denoted as F_{seq} . (2) Additionally, the process yields row attention maps, which are designated as $Attent$.

The MSA Transformer is constructed with an architecture comprising twelve encoders, each possessing twelve attention heads, culminating in a total of 144 attention maps organized in a $c \times c \times 144$ formation. The row attention map derived from this setup, along with the initial input features, are merged along the feature dimension to produce a 2D feature map. This resulting feature map, encapsulated as a tensor of dimensions $c \times c \times 400$, is termed the MSA feature. Subsequently, the feature dimension of the MSA feature is condensed to 64 through the application of three 2D convolutional layers. Each convolutional layer is succeeded by instance normalization and activation through a Rectified Linear Unit (ReLU). This resultant MSA feature, designated as F_{MSA} , emerges from the MSA Transformer's processing sequence as outlined.

$$\begin{aligned} F_{seq, Attent} &= \text{Transformer}(MSA), \\ F_{inter} &= F_{seq} \oplus Attent, \\ F_{MSA} &= \text{ConvBlock}(F_{inter}). \end{aligned}$$

3.2.2. PSSM feature generation

In this study, we construct the PSSM for each protein sequence by conducting four rounds of the PSI-BLAST analysis [24], utilizing the Uniprot_sprot database [41] with its standard settings. The produced PSSM is formatted as a matrix of dimensions $c \times n$, where c denotes the protein sequence length and n represents the count of standard amino acid residues. Each entry in the PSSM quantifies the position-specific affinity of the 20 amino acids at a certain sequence position, with the scores usually manifesting as positive or negative integers. These scores are indicative of the evolutionary probability that an amino acid at a specific position might evolve into another amino acid characterized by a superior score, thereby suggesting a propensity for a more stable structural conformation.

3.2.3. HMM feature generation

In this research, the HMM profiles are produced utilizing HMMER version 3.3.2 [37], applying both the default parameters and the standard sequence database. We characterize the resultant HMM profile as a matrix with dimensions $c \times n$, where c represents the length of the protein sequences in question, and n denotes the count of standard amino acids. The HMM matrix typically exhibits scores as positive non-integer values. Serving as a stochastic model, the HMM plays a crucial role in the prediction of protein secondary structures, with the explicit sequence being the amino acid sequence and the hidden state corresponding to their secondary structure configurations. Through training, an HMM generates probabilistic transition matrices for each state, thus delineating various protein families. Within the context of PSSP, HMMs are employed to discern the patterns of H, E, and C structures, facilitating their application to novel protein sequences whose secondary structures are yet to be determined.

3.2.4. Sequence embedding feature generation

In this study, we utilize the word2vec methodology [42] to extract sequential features from protein sequences. The input for this process is conceptualized as an $r \times c$ matrix, where r stands for the batch size, and c denotes the length of individual protein sequences. The resultant output from word2vec is formulated as an $r \times c \times \text{embedding_dim}$ tensor, with embedding_dim indicating the embedding vector's dimensionality, essentially reflecting the number of dimensions allocated for the representation of each symbol. When we use word2vec to extract the embedding features, the embedding layer is set to $\text{num_embeddings} = 25$, $\text{embedding_dim} = 20$. num_embeddings represents the number of words in the dictionary, 22 of them represent the amino acid residues ('A', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'K', 'L', 'M', 'N', 'P', 'Q', 'R', 'S', 'T', 'V', 'W', 'Y', 'X', 'NoSeq'), 3 of them represent the 3-state secondary structure label. embedding_dim represents the dimension of the embedding vector, which we want to correspond to the 20 amino acids in a real protein, so we set it to 20. Applying the Word2vec model, our method can learn high-quality residual vectors with latent semantics and prevent zero-redundant representations.

3.2.5. Fusion strategy

Here we use the cosine similarity-based feature fusion strategy [38] to generate a unified feature representation space by fusing the above four kinds of features. We define the cosine similarity-based feature as similarity information generated by two input features, which have uniform feature representation spaces. The cosine similarity-based feature is calculated as follows:

$$\begin{aligned} \text{cosine}(X, Y) &= \frac{X \cdot Y}{\|X\|_{\infty} \|Y\|_{\infty}}, \\ \|X\|_{\infty} &= \max_{1 \leq i \leq n} \sum_{j=1}^n |x_{ij}|, \end{aligned}$$

where $\|X\|_{\infty}$ and $\|Y\|_{\infty}$ is the infinite-norm of input feature X and Y .

With this, we can calculate the fused feature for each given protein sequence, which is denoted as a matrix F_{fuse} as below:

$$F_{\text{fuse}} = P + \cosine(P, E) + \cosine(H, E) + H + F_{\text{MSA}},$$

where F_{MSA} represents MSA feature, P represents PSSM feature, H represents HHM feature and E represents sequence embedding feature. When the cosine similarity-based feature fusion strategy is used in the feature extraction stage, it brings many advantages that are worthy of attention in the field of deep learning. Firstly, cosine similarity, as an effective measure, can evaluate the similarity between features in the feature space, which helps to find the complementarity between different features. Through this method, the features with similarity can be fused, to integrate the feature representations from different information sources, and enhance the diversity and integrity of the features. Secondly, this strategy helps to eliminate duplicate information in the features, reduces redundancy, effectively reduces the risk of overfitting, and improves the generalization ability of the model. So in Fig. 2, the colored numbers 1 to 5 in the Multi-feature fusion module represent P , $\cosine(P, E)$, $\cosine(H, E)$, H , and F_{MSA} , respectively. Finally, four kinds of features are concatenated to a 2D matrix as the high-latent input feature for the following hybrid neural network.

3.3. Transformer encoder & decoder

In our study, we leverage a transformer network equipped with an attention mechanism, a technique that has shown remarkable effectiveness in the domain of natural language translation, to identify relevant sequence contexts throughout the entire protein sequence. This approach aims at predicting the secondary structure type for each amino acid position. Such a deep learning framework is adept at capturing long-distance interactions between amino acids, analogous to word relationships within an English sentence, which are crucial for accurate secondary structure prediction.

Our specifically tailored model, MFTrans, utilizes an encoder-decoder architecture for translating protein sequences into secondary structures. The encoder function transforms an input feature representing a protein in symbolic form into a series of continuous values (internal features), with dimensions $L \times 512$ (where L denotes the length of the protein sequence). Subsequently, based on these internal features, the decoder generates a sequence representing the 3-state protein secondary structures sequentially in an auto-regressive manner, thus enabling precise structure prediction.

Given the fusion feature as input, MFTrans employs an encoder-decoder process to infer the protein's secondary structure. The encoder comprises an input embedding component and a self-attention component. Initially, the input embedding component converts the input tokens into value vectors of dimensions $L \times L \times 64$. Subsequently, the input embedding vector undergoes processing by the self-attention component, which comprises a stack of $N = 6$ identical sub-components. Each sub-component entails a multi-head self-attention layer (with a number of heads = 8) and a fully connected feed-forward layer. Additionally, each layer incorporates a residual connection, followed by layer normalization and a dropout operation with a dropout rate of 0.1.

Similarly structured, the decoder comprises an embedding component and two attention components. It commences with the output tokens (i.e., true 3-state secondary structure during training or predicted secondary structure during inference) from previous positions in the target protein, which are then converted into embedding vectors of dimensions $L \times 512$. A sinusoidal position encoding vector is added atop the embedding vector, after which it enters the first self-attention component, followed by the cross-attention component. The first self-attention component, akin to the encoder, comprises a stack of $N = 6$ identical layers, with the distinction of applying a masked self-attention mechanism to ensure that predictions for the current position solely rely

on preceding output tokens during training. In the inference phase, the decoder's self-attention component considers only previously generated secondary structure states when generating the subsequent ones. Subsequently, both the output of the decoder's self-attention component and the output of the encoder stack feed into the cross-attention decoder component, which is composed of $N = 6$ identical sub-components. Each sub-component encompasses a multi-head self-attention layer (with a number of heads = 8) and a fully connected feed-forward layer. Finally, the output of the decoder stack is transformed into the 3-state PSSP via linear transformation and softmax function.

3.4. Hybrid neural network

The MFTrans model integrates a hybrid neural network structure, combining CNN and BiGRU layers for feature extraction. This network is designed to process high-latent features, utilizing CNN for local feature extraction and BiGRU for global contextual insights. Initially, the fused features from the preceding module are fed into a CNN architecture, which includes a sequence of layers: a 2D convolutional layer for spatial feature extraction, a batch normalization layer for stabilizing learning, a max pooling layer for dimensionality reduction, and a ReLU activation function layer for non-linear transformation. The CNN's role is to capture local patterns within the protein feature matrix, as each convolutional filter focuses on a specific receptive field. To enhance training efficiency and model generalization, a dropout layer is applied prior to the CNN operation. The features produced by the CNN, denoted as F' , and the previously fused features, F_{fuse} , undergo processing in the following manner:

$$F' = \text{CNN}(\text{Dropout}(F_{\text{fuse}})).$$

For the purpose of deriving a comprehensive global feature depiction, the features denoted as F are processed through BiGRU layers. The features that have been successfully extracted in this manner are represented by F_{extract} . This extraction process involves the deployment of two separate BiGRU layers, which are tasked with extracting features both from the CNN-derived data and the initial protein embeddings, as outlined below:

$$F_{\text{extract}} = \text{BiGRU}_2(\text{BiGRU}_1(F')).$$

The process of extracting global features is segmented into two distinct phases, delineated by a mean squared error (MSE) restriction, details of which will be discussed subsequently. In the initial phase of global feature extraction, the first BiGRU layer functions as an intermediary, succeeding the local feature extraction stage. This layer processes an enhanced feature matrix to further elucidate long-range dependencies within the data. The subsequent phase involves the second BiGRU layer, which is then succeeded by a series of FC layers, facilitating a deeper level of feature integration and extraction.

3.4.1. MSE restriction

Incorporated between the two BiGRU layers dedicated to the extraction of global features, we implement a MSE constraint [43] to facilitate the model's learning of global sequence features more effectively. This MSE constraint involves FNN layers that process the output from the initial stage of global feature extraction. The output from the FNN layer is denoted as F_{ref} . This setup engages in a dual-function approach: firstly, it utilizes the secondary structure labels alongside F_{ref} to compute the MSE; secondly, it determines the loss by comparing the actual labels with the model's predictions. The loss incurred from the MSE constraint is symbolized as L_{restrict} . To compute the final outcome, these two components are aggregated as described:

$$\begin{aligned}
F_{\text{ref}} &= FNN(\text{BiGRU}_1(F)), \\
L_1 &= \text{Loss}_{\text{MSE}}(\text{Predict}, \text{Label}), \\
L_2 &= \text{Loss}_{\text{MSE}}(F_{\text{ref}}, \text{Label}), \\
L_{\text{restrict}} &= L_1 + \lambda L_2,
\end{aligned}$$

where λ is a balance coefficient used to balance the two loss functions L_1 and L_2 to calculate L_{restrict} .

3.5. Secondary structure prediction

Within biological organisms, twenty standard amino acids serve as the essential building blocks for protein sequences. In the context of a 3-state PSSP task, each amino acid x_i within a sequence is associated with a label y_i from the set H, E, C, thereby generating a sequence reflective of the protein's secondary structure. Following the processing through a Transformer encoder, three FC layers are utilized. These layers are designed to intake the refined features and subsequently produce outputs corresponding to the secondary structure labels as delineated:

$$\text{Predict} = FNN(F_{\text{enhance}}),$$

where FNN symbolizes a series of three consecutive FC layers, which serve to diminish the dimensionality of the feature from 1024 to 3. The resultant prediction is encapsulated as an $r \times c \times 3$ tensor, wherein the secondary structure labels are delineated through one-hot encoding. To enhance the robustness of our model, we formulate a specialized loss function for the prediction module. This loss function is comprised of two distinct cost components:

$$\begin{aligned}
L_3 &= \frac{1}{r} \sum_{i=1}^r \sum_{j=1}^c (Z_i^j - W_i^j), \\
L_{\text{total}} &= \gamma L_{\text{restrict}} + L_3,
\end{aligned}$$

where L_3 represents the MSE loss between feature extraction and prediction. r is the number of input sequences and c is the length of each input sequence. In fact, r is the batch size. During training and testing, we set $r = 1$, which means the model predicts the secondary structure of a single amino acid sequence. Z represents the extracted feature generated by the feature extraction module. W is defined as the weighted secondary structure label, which will be discussed in implementation. L_{restrict} represents the loss calculated by the MSE restriction. γ is a balance coefficient used to balance the two loss functions to calculate the final cost for the optimizer.

4. Experiment

4.1. Dataset

For an equitable evaluation against contemporary cutting-edge approaches, our MFTrans framework undergoes training and validation on the CullPDB [44] dataset's designated training and validation subsets. We utilize the datasets CB513 [45], CASP12, CASP13, CASP14 [46], TEST2016 and TEST2018 [47] as benchmarks to assess our

Table 1
Details of datasets used in this work.

Datasets	Structural States			Number
	H	E	C	
Train set	183,429	102,159	173,261	5600
Val set	20,564	12,030	19,751	256
CB513	29,078	19,103	35,973	514
CASP12	875	624	968	11
CASP13	898	913	1568	13
CASP14	326	278	476	9
TEST2016	108,968	65,669	113,230	1212
TEST2018	19,122	11,692	20,045	250

methodology's effectiveness and to facilitate comparative analysis with preceding strategies. The datasets referenced are summarized in Table 1.

Within the CullPDB dataset, a total of 6133 proteins were harvested from the CullPDB server, adhering to specific criteria: a resolution finer than 2.5 Å, R-factor < 1.0, and sequence identity < 25 % for any pair of proteins as determined by BlastClust [24]. Following the removal of redundancies with CB513, the dataset is segmented into 5600 training instances, 256 for validation, and 272 for testing. The composition of the training dataset is depicted in Fig. 3. Part a of this figure illustrates the distribution of protein sequence lengths alongside the prevalence of H, E, and C structures within each sequence. Part b of the figure delineates the frequency of protein sequences corresponding to the specific counts of each structural state present in the dataset. Our investigation exclusively utilizes the training and validation subsets derived from the CullPDB dataset. CB513 [45], recognized as a benchmark dataset of significant repute and widespread application, is integrated as one of our testing sets subsequent to the adjustment for sequence similarities between CULLPDB and CB513.

The critical assessment of protein structure prediction (CASP) initiative, launched in 1994, represents a biannual contest aimed at fostering advancements in the prediction of protein structures based solely on their amino acid sequences. This event is a collective endeavor by the scientific community to push the boundaries of protein structure modeling [46]. From the array of datasets generated through the CASP competitions, we have selected CASP12, CASP13, and CASP14 for our testing purposes, attributing to their widespread acceptance and utilization. Additionally, the TEST2016 and TEST2018 dataset, curated by Hanson et al. [47], encompasses 1212 and 250 unique, high-quality protein structures that were publicly disclosed in the Protein Data Bank (PDB). To ensure the integrity of our evaluation and mitigate any bias stemming from overlapping data, we ensured that all test datasets exhibit <25 % sequence similarity with our training and validation sets, as verified by BlastClust.

4.2. Implementation details

Our model construction utilizes PyTorch [48]. The multi-feature fusion module incorporates a pre-trained MSA Transformer. To enhance the training efficiency and performance of our network, we implement batch normalization and dropout strategies. Specifically, we incorporate a dropout layer with a 0.5 rejection rate prior to the CNN layer. Batch normalization is applied in three critical areas: 1) within the CNN, 2) at the juncture between the BiGRUs, and 3) amidst the transformer encoder and decoder, aiding in the mitigation of internal covariate shift by managing the mean and variance within the network. In terms of regularization, we set the MSE constraint parameter λ and the balance coefficient γ in the overall loss equation to 0.1. To address the variance in residue numbers across the three structural states, we employ a weighted secondary structure label for calculating Loss_3 in L_{total} , assigning weights of 1.25 to state E and 1 to states H and C, respectively. Throughout the training period, the MSA Transformer's parameters are kept constant, whereas the rest of the MFTrans parameters are optimized using the Adam optimizer at a learning rate of 1e-3. The training, conducted on an NVIDIA GeForce RTX A6000 GPU, spanned approximately 100 epochs, culminating in a duration of around 10 days.

During the training phase, we implement a subsampling approach for handling MSA, which fulfills dual objectives: it acts as a method of data augmentation to enhance model robustness and prevents GPU memory from being overwhelmed by large MSAs. To reduce computational demands and avert potential declines in model efficacy, our strategy involves not supplying the full MSA as input. Rather, we opt for a subset of MSA that is computationally manageable and aligns with the training input size. This involves randomly selecting rows from the MSA, setting a cap at $2^{14}/c$ rows (where c denotes sequence length) and ensuring a minimum selection of 16 rows. Importantly, we always

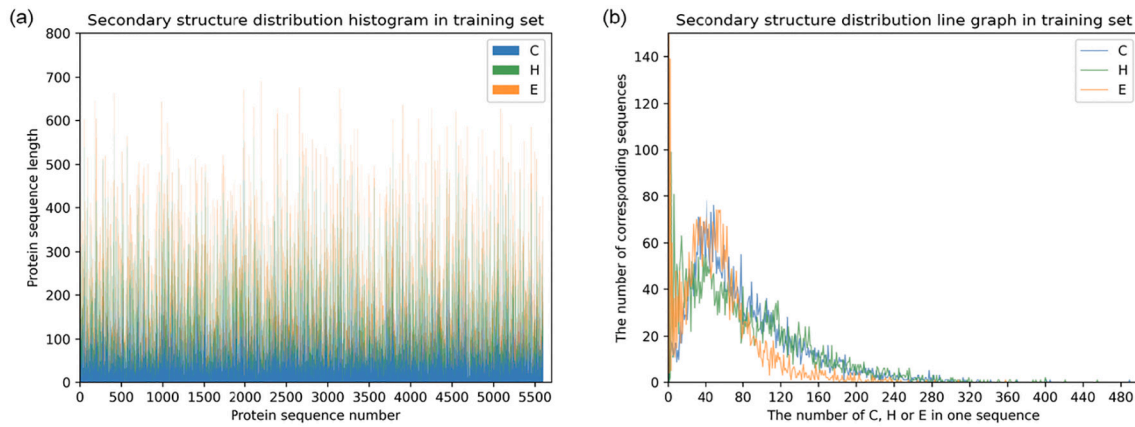


Fig. 3. Statistics of training dataset. a) The number of each secondary structure in each protein sequence. b) The number of corresponding protein sequences in response to the specified amount of C, H or E.

include the query sequences in the first row. For the inference phase, we refine our subsampling to include 256 sequences by incorporating the sequence that exhibits the lowest average Hamming distance.

4.3. Performance metrics

In our experimental work, we employed accuracy as the metric for evaluating MFTrans's performance. Here, *ACC* signifies the overall predictive accuracy across all conditions. The accuracy specific to each structural condition is denoted as ACC_i where i stands for *H*, *E*, *C*. The computation methods for these indicators are outlined below:

$$ACC_i = \frac{A'_i}{A_i}, ACC = \frac{\sum_{i \in \{H, E, C\}} A'_i}{\sum_{i \in \{H, E, C\}} A_i},$$

where A_i is the total number of existing residues in each state and A'_i is the number of correctly predicted residues in the state i .

5. Results

5.1. Comparison experiment

We conduct a comparative analysis of our approach against established state-of-the-art (SOTA) techniques for PSSP across various datasets. The outcomes on the CB513 dataset are presented in Table 2. The evaluation of performance, as depicted in Tables 2–6, utilizes the Q3 accuracy metric. This metric encompasses a tri-class categorization, comprising alpha-helix (H), beta-strand (E), and coil (C) structures. Q3 accuracy represents the proportion of protein sequence positions where the secondary structure is correctly identified. The discrepancy, also known as the error or cost, measures the divergence between the predicted outcomes of the model and the actual results. For each amino acid residue, we compute the MSE between the model's predictions and the

Table 2

The performance comparison between MFTrans and other SOTA models on the CB513 dataset. Bold corresponds to best performance for each metric, and underline indicates the second best.

Methods	ACC
ProtT5-XL-BFD	0.8400
ProtBert-BFD	0.8300
MUFold-SS	<u>0.8512</u>
ReportX	0.8234
Spider3	0.8456
MFTrans(Ours)	0.8834

Table 3

The performance comparison between MFTrans and other SOTA models on the CASP datasets.

Datasets	Methods	ACC
CASP12	ProtT5-XL-UniRef50	0.8100
	MUFOLD-SS	0.8059
	DeepCNF-SS	<u>0.8131</u>
	PSIPRED	0.8051
	MFTrans(Ours)	0.8483
CASP13	DeepCNF-SS	0.7780
	PSIPRED	<u>0.7980</u>
	Nnessy	0.7570
	Sspro	0.7520
CASP14	MFTrans(Ours)	0.8247
	NetSurfP-2.0	0.7539
	OPUS-TASS	0.7730
	OPUS-TASS2	<u>0.8087</u>
	SPOT-1D	0.7519
	MFTrans(Ours)	0.8431

Table 4

The performance comparison between MFTrans and other SOTA models on the TEST2016 dataset.

Methods	ACC
SPOT-1D-single	0.7513
NetSurfP-2.0	0.8345
OPUS-TASS	0.8214
MUFOLD	<u>0.8538</u>
SPOT-1D	0.8146
MFTrans(Ours)	0.8717

Table 5

The performance comparison between MFTrans and other SOTA models on the TEST2018 dataset.

Methods	ACC
SPOT-1D-single	0.7428
NetSurfP-2.0	0.8531
OPUS-TASS	<u>0.8684</u>
MUFOLD	0.8478
SPOT-1D	0.8618
MFTrans(Ours)	0.8842

expected outcomes. The elaboration on loss functions is provided in Section 3.4.1.

Within the CB513 dataset, our model establishes a new benchmark, attaining a SOTA accuracy (ACC) of 0.8834. Demonstrating the efficacy

Table 6

Results of ablation experiments on CB513 dataset. The first four lines means models lacking corresponding input feature, 'Full' means the complete MFTrans.

Methods	ACC_H	ACC_E	ACC_C	ACC
W/o MSA feature	0.8735	0.7345	0.7562	0.7921
W/o PSSM feature	0.8865	0.7862	0.8064	0.8236
W/o HMM feature	<u>0.8964</u>	<u>0.7932</u>	<u>0.8365</u>	<u>0.8462</u>
W/o sequence feature	0.8169	0.6862	0.7026	0.7462
Full	0.9316	0.8378	0.8593	0.8834

of the SOTA MFTrans method in ACC metrics, it secures a 3 % enhancement over prior SOTA methodologies. The integration of multi-feature fusion and a hybrid neural network architecture in our approach significantly bolsters PSSP accuracy. Consistent superior performance is observed in evaluations conducted on the CASP datasets, TEST2016 and TEST2018, as detailed in Tables 3, 4 and 5. On CASP12, CASP13, and CASP14 datasets, MFTrans outperforms, achieving ACC values of 0.8483, 0.8247, and 0.8431, respectively. Furthermore, MFTrans achieves ACC values of 0.8717 in TEST2016, and ranks foremost in TEST2018, with an ACC of 0.8842. Collectively, across the CASP series, TEST2016 and TEST2018, our outcomes surpass those of previous SOTA techniques by approximately 3 % in ACC metrics.

AlphaFold, recognized for its pivotal role in protein structure prediction, serves as a benchmark within the industry. By juxtaposing our model with AlphaFold, we confirmed our model's efficacy and gauged its precision, thereby bolstering the credibility and dependability of our investigative efforts. We enhanced the AlphaFold network by integrating mean pooling and Max pooling techniques at its penultimate layer, followed by appending an extra one-dimensional output segment dedicated to forecasting secondary structure classifications. In CASP14, AlphaFold's network achieved a Q3 prediction accuracy of 83 %, marginally surpassing our model's accuracy by approximately 1 %. Notably, AlphaFold's training regimen demands the utilization of 128 TPUs over an 11-day period. In contrast, our model achieves comparable experimental outcomes with merely 10 days of training on a single NVIDIA GeForce RTX A6000 GPU, utilizing just 25 % of the memory required by AlphaFold. Moving forward, our focus will remain on refining the prediction precision of our model, leveraging its efficiency and swift inference capabilities.

5.2. Ablation study

Through meticulously designed ablation studies within the MFTrans framework, we elucidate the impact of each individual feature on the overall performance. These experiments, conducted on the CB513 dataset, aim to pinpoint the significance of various input features by evaluating the model's efficiency in the absence of each: namely, the MSA feature, the PSSM feature, the HMM feature, and the sequence embedding feature. The findings from these ablation tests are systematically tabulated in Table 6, revealing nuanced insights into the contributions of each feature towards the model's predictive accuracy.

The empirical evidence underscores a notably diminished performance of the MFTrans model in scenarios lacking the sequence feature, underscoring the pivotal role of sequence embeddings in the fusion of multi-features. This degradation highlights the intrinsic value of sequence data in capturing the fundamental characteristics necessary for accurate secondary structure prediction. Furthermore, the complete incarnation of MFTrans, encompassing all proposed features, consistently outperforms its variants with omitted features across all evaluated metrics. This conclusively affirms the integral necessity and synergistic effectiveness of each included feature within the MFTrans framework, bolstering the model's robustness and precision in predicting protein secondary structures. The comprehensive inclusion of diverse features thus stands validated as a cornerstone for achieving superior results, reinforcing the innovative architecture of MFTrans as a groundbreaking

approach in the domain of protein structure prediction.

In a further exploration to validate the critical contributions of distinct components within the MFTrans framework, we executed an additional set of ablation studies, specifically focusing on the architecture's components. This examination was carried out using the CB513 dataset, with findings meticulously documented in Table 7. Recognizing the multifaceted design of our approach, we scrutinized the performance implications of omitting key architectural elements: the CNN layers, the BiGRU layers, and the Transformer component.

The outcomes from these targeted ablation tests revealed a decrement in MFTrans's predictive accuracy when any of the mentioned components was excluded from the model configuration, signifying their integral roles. Notably, the removal of CNN layers resulted in the most pronounced accuracy decline, underscoring the CNN's pivotal function in extracting spatially relevant features from protein sequences. This observation underscores the essentiality of CNN layers in capturing complex patterns and details critical for accurate secondary structure prediction.

Consequently, it becomes evident that the complete MFTrans architecture, with its comprehensive incorporation of CNN, BiGRU, and Transformer components, achieves the highest performance benchmarks. This finding reinforces the premise that each designed component of MFTrans is indispensable, contributing uniquely to the model's overall efficacy. The synergistic integration of these components underlines the architecture's robustness and its innovative capacity to accurately predict protein secondary structures, highlighting the significance of each element in the ensemble's effectiveness.

In addition to exploring the architectural components and features, we also delved into the optimization of hyper-parameters within the MFTrans framework, specifically focusing on the MSE constraint to ascertain its impact on the model's prediction accuracy. This investigation entailed conducting ablation experiments to discern the performance variance attributable to modifications in a single parameter, thereby isolating the influence of said parameter on the overall system efficacy. The outcomes of these experiments, which were performed on a validation dataset, are encapsulated in Fig. 4.

The empirical findings from this meticulous examination indicated an optimal performance of the MFTrans model on the validation set when the hyper-parameters λ and γ were both adjusted to 0.1. This configuration suggests that setting λ , the parameter governing the strength of the MSE constraint, to 0.1, alongside a balance coefficient γ also set at 0.1 within the total loss calculation, facilitates the most favorable outcome in terms of prediction accuracy during the training phase.

Such a detailed parameter analysis underscores the criticality of fine-tuning hyper-parameters in enhancing the predictive capabilities of DL models, specifically within the context of PSSP. The optimization of λ and γ not only aligns with the goal of achieving high accuracy but also reflects the nuanced balance required in loss function composition to guide the model towards effective learning outcomes. Consequently, these insights provide valuable guidance for setting hyper-parameters in the MSE restriction, affirming their substantial role in refining the MFTrans model's performance and contributing to its advanced prediction accuracy.

Table 7

Results of ablation experiments on CB513 dataset. The first four lines means models lacking different components, 'Full' means the complete MFTrans.

Method	ACC_H	ACC_E	ACC_C	ACC
W/o CNN Layers	0.7615	0.7185	0.7560	0.7421
W/o Transformer	0.8134	0.7256	0.7924	0.7843
W/o BiGRU Layers	<u>0.8531</u>	<u>0.7924</u>	<u>0.8434</u>	<u>0.8315</u>
Full	0.9316	0.8378	0.8593	0.8834



Fig. 4. Results of ablation experiments on validation dataset. The X-axis represents the value of λ , while the Y-axis represents the value of γ , the tag indicates the accuracy on the validation set.

5.3. Visualization of performance comparison

To vividly showcase the comparative effectiveness of MFTrans against established techniques, we employ PyMOL [49] for the visual mapping of secondary structures onto their corresponding tertiary configurations. We conduct a comparative analysis by randomly selecting two proteins, identified by their PDB IDs, to assess the prediction accuracies of various methods concerning their secondary structures. In the visual representations provided in Fig. 5, we examine the performance of MFTrans alongside OPUS-TASS and MUFold-SS on the proteins designated by two proteins (PDB ID: 1KY3 and 2FD3). The visualization distinctly indicates that MFTrans's predictions more closely align with the ground truth structures when compared to the other methodologies. This outcome substantiates the superior predictive capability of MFTrans over the current alternatives, emphasizing its enhanced accuracy and potential to revolutionize the field of PSSP through more precise and reliable modeling.

5.4. Case study

To assess the precision of our model's predictions at mutation sites, we conduct experiments utilizing SARS-CoV-2 proteins and employ our

MFTrans for secondary structure prediction. Severe acute respiratory syndrome coronavirus 2(SARS-CoV-2) infection poses a great threat to global public health, and predicting the evolution direction of the virus is of great significance for the prevention and control of related diseases. The mutation rate of this virus is higher than that of double-stranded DNA viruses, indicating strong evolutionary potential. In addition, the presence of a large number of post-translational isozyme products of SARS-CoV-2 suggests rapid mutation and strong host adaptation.

SARS-CoV-2 virus comprises four structural proteins: spike (S), envelope (E), membrane (M), and nucleocapsid (N) proteins [50]. Notably, the spike protein encompasses two subunits - S1 and S2 - and assumes a pivotal role in interacting with host cell receptors, thereby emerging as a principal target for the development of COVID-19 vaccines. Predicting and analyzing the secondary structure of the mutated spike protein can provide a deep understanding of the mutation characteristics of the virus, lay a foundation for the design of effective drug therapy, and provide a basis for subsequent epidemic prevention and control and epidemiological studies.

We collect the sequencing results of several SARS-CoV-2 viruses published in the PDB database, searching for their high frequency mutation sites, and using MFTrans to predict their secondary structure sequentially. We opt to analyze the BA.2 protein and XBB protein alongside several representative proteins from these new variants to conduct secondary structure prediction. Illustrated in Fig. 6, we showcase the predicted structures generated by our method. Within the visualization, the blue area indicates incorrect predictions, whereas the red area denotes correct predictions. Furthermore, to emphasize the prediction outcomes at mutation sites, non-mutation sites are rendered transparent. Experimental results reveal an accuracy rate of 90.42 % in mutation areas and an overall prediction accuracy rate of 88.43 %. These findings underscore the robust performance of MFTrans, particularly in accurately predicting secondary structures at mutation sites.

5.5. Discussion

Compared to other models, our model shows superior performance on public data sets. This may be attributed to its ability to capture multi-view information in secondary structure predictions. Our model combines multiple deep learning modules to make predictions, improving the accuracy of predictions. Despite the encouraging outcomes, our

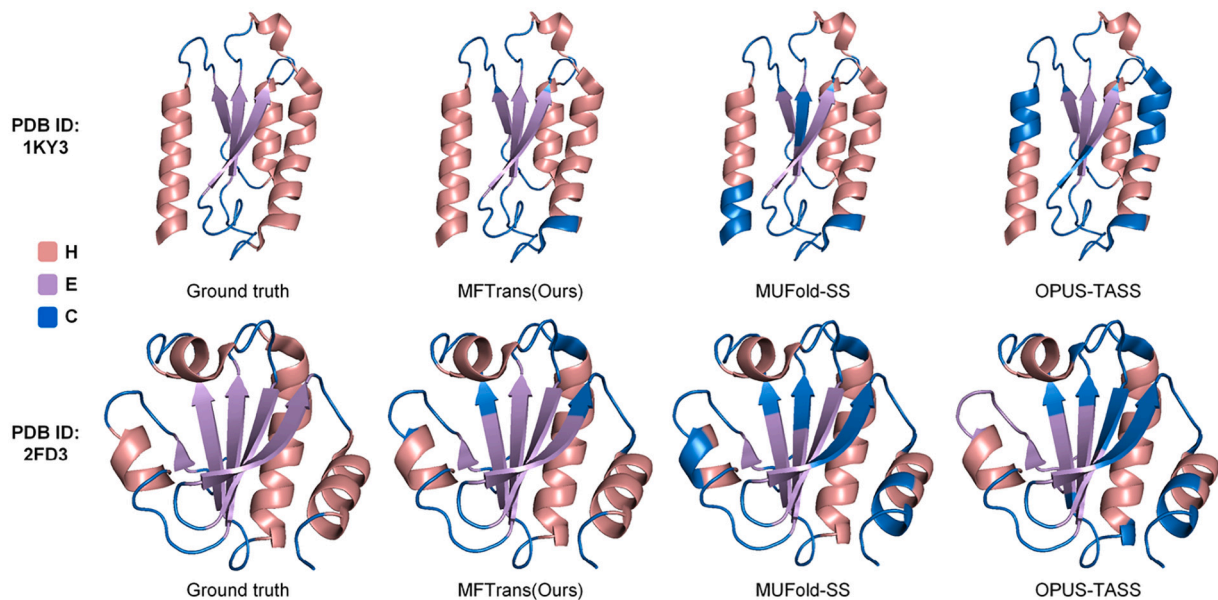


Fig. 5. Visualization of secondary structures mapped into tertiary structures for MFTrans and other SOTA methods. Areas colored in red, purple, and blue denote H, E, and C, respectively.

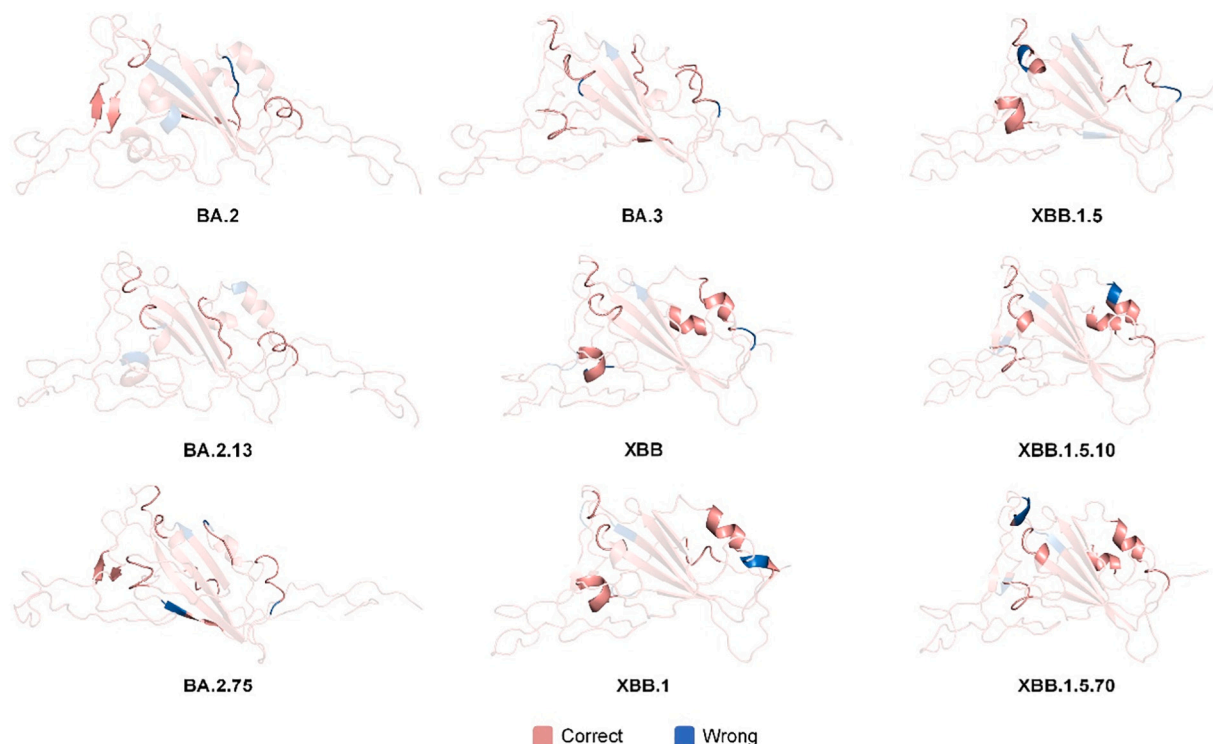


Fig. 6. Visualization of prediction results for MFTrans on SARS-CoV-2 proteins. The blue area represents incorrect predictions, while the red area represents correct predictions. Non-mutation sites are displayed as transparent to emphasize the prediction results at the mutation sites.

model presents certain limitations. Our model requires a lot of data preprocessing time, since the proposed method relies on the MSA input, which requires a time-consuming database search process.

The different features used in our method do provide important information for prediction and have been widely applied in other protein prediction fields. However, the feature is mainly derived from homology sequences. In the absence of useful information such as homologous sequences, the model may face many challenges, which should be focused on and solved in our research. We conducted a preliminary study on this, and the preliminary results showed that there may be a degradation in model performance in the absence of homologous sequence information.

In our future work, the use of distillation learning is a compelling solution for low-homology PSSM situations. Using pre-trained protein sequence language models through distillers can provide important directions for future research. This method is expected to improve the performance of the model in the face of the challenge of low homology and provide a new way to solve the current problem of insufficient prediction accuracy. With the development of the field of protein structure prediction, the method of distillation learning may bring breakthroughs for the improvement of prediction models.

In addition, the size of the dataset, while sufficient for supervised feature extraction, is still relatively limited and directly affects the performance of the model. Future computational efforts should prioritize the combination of expanded datasets and more comprehensive biological knowledge.

6. Conclusion

In our research, we introduced MFTrans, a deep learning-focused multi-feature fusion network tailored for PSSP, which demonstrated outstanding efficacy in delineating proteins' secondary structures. By amalgamating diverse feature representations including MSA, PSSM, HMM, and sequence embeddings, our multi-feature fusion approach significantly bolstered the representational richness of features, thereby

enhancing the predictive precision of our model. MFTrans is distinguished by its innovative hybrid network architecture, which merges CNN and BiGRU networks to meticulously extract both the local and global attributes of protein sequences, underpinning our model's high performance. Through comprehensive ablation studies, we ascertained the indispensable contribution of each MFTrans component towards achieving exceptional accuracy, underscoring the efficacy of our proposed strategy. Crucially, MFTrans has surpassed existing methodologies across various datasets, marking a notable advancement in PSSP research.

Moreover, MFTrans exhibits the capability to incorporate further structural data about proteins, paving the way for a more effective exploration of protein architectures, thereby deepening our grasp of their biological roles. The ramifications of our investigation extend across several scientific domains, such as pharmaceutical development, diagnostic methodologies for illnesses, and the engineering of proteins. The methodological prowess of MFTrans, characterized by its integration of various feature representations and its adept extraction of both macroscopic and microscopic protein traits, significantly enriches our comprehension of protein structure and functionality. In comparison to other models dedicated to PSSP, MFTrans stands out for its superior performance, with its predictions more closely aligning with the empirical truth. This enhancement not only showcases the technical advancements of MFTrans but also its potential to contribute substantially to the fields of bioinformatics and molecular biology, among others, by facilitating more accurate predictions that are critical for forward-looking research endeavors in drug discovery, disease understanding, and the customization of protein properties for specific applications.

In summary, MFTrans introduces a novel paradigm for PSSP, with the added advantage of potentially incorporating more comprehensive structural data on proteins. This approach facilitates a deeper insight into the biological functions of proteins, bearing profound implications for areas such as drug discovery, disease diagnosis, and protein engineering. MFTrans represents a promising methodology that not only

enhances prediction accuracy but also opens new avenues for research, contributing to the advancement of novel therapeutic and diagnostic solutions within the realm of protein structure prediction. The capability of MFTrans to amalgamate multiple feature representations and accurately extract both broad and detailed features of protein sequences positions it as a significant stride forward in the computational biology field, promising to accelerate scientific breakthroughs and practical applications in healthcare and biotechnology.

CRedit authorship contribution statement

Yifu Chen: Data curation, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Guanxing Chen:** Data curation, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Calvin Yu-Chian Chen:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this report.

Data availability

The source code and data of this study are available at <https://github.com/chenyf96/MFTrans>.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant No. 62176272), Research and Development Program of Guangzhou Science and Technology Bureau (2023B01J1016), and Key-Area Research and Development Program of Guangdong Province (No. 2020B1111100001).

References

- [1] X. Fang, F. Wang, L. Liu, J. He, D. Lin, Y. Xiang, K. Zhu, X. Zhang, H. Wu, H. Li, et al., A method for multiple-sequence-alignment-free protein structure prediction using a protein language model, *Nat. Mach. Intell.* 5 (2023) 1087–1096.
- [2] W. Zheng, Q. Wuyun, Y. Li, C. Zhang, P.L. Freddolino, Y. Zhang, Improving deep learning protein monomer and complex structure prediction using DeepMSA2 with huge metagenomics data, *Nat. Methods* 1–11 (2024).
- [3] K.E. Wu, K.K. Yang, R. van den Berg, S. Alamdari, J.Y. Zou, A.X. Lu, A.P. Amini, Protein structure generation via folding diffusion, *Nat. Commun.* 15 (2024) 1059.
- [4] E.F. McDonald, T. Jones, L. Plate, J. Meiler, A. Gulsevin, Benchmarking AlphaFold2 on peptide structure prediction, *Structure* 31 (2023) 111–119.
- [5] K. Harini, M. Sekijima, M.M. Gromiha, PRA-Pred: structure-based prediction of protein-RNA binding affinity, *Int. J. Biol. Macromol.* 259 (2024) 129490.
- [6] J. Garnier, D.J. Osguthorpe, B. Robson, Analysis of the accuracy and implications of simple methods for predicting the secondary structure of globular proteins, *J. Mol. Biol.* 120 (1978) 97–120.
- [7] C.A. Goverde, B. Wolf, H. Khakzad, S. Rosset, B.E. Correia, De novo protein design by inversion of the AlphaFold structure prediction network, *Protein Sci.* 32 (2023) e4653.
- [8] Y. Jiang, R. Wang, J. Feng, J. Jin, S. Liang, Z. Li, Y. Yu, A. Ma, R. Su, Q. Zou, et al., Explainable deep hypergraph learning modeling the peptide secondary structure prediction, *Adv. Sci.* 10 (2023) 2206151.
- [9] G. Terashi, X. Wang, D. Prasad, T. Nakamura, D. Kihara, Deepmainmast: integrated protocol of protein structure modeling for cryo-em with deep learning and structure prediction, *Nat. Methods* 21 (2024) 122–131.
- [10] J. Hanson, K. Paliwal, T. Litfin, Y. Yang, Y. Zhou, Improving prediction of protein secondary structure, backbone angles, solvent accessibility and contact numbers by using predicted contact maps and an ensemble of recurrent and residual convolutional neural networks, *Bioinformatics* 35 (2019) 2403–2410.
- [11] S. Wang, J. Peng, J. Ma, J. Xu, Protein secondary structure prediction using deep convolutional neural fields, *Sci. Rep.* 6 (2016) 1–11.
- [12] C. Fang, Y. Shang, D. Xu, MUFOLD-SS: new deep inception-inside-inception networks for protein secondary structure prediction, *Proteins Struct. Funct. Bioinforma.* 86 (2018) 592–598.
- [13] Y. Ma, Y. Liu, J. Cheng, Protein secondary structure prediction based on data partition and semi-random subspace method, *Sci. Rep.* 8 (2018) 1–10.
- [14] Y. Liu, Y. Ma, J. Cheng, A novel group template pattern classifiers (GTPCs) method in protein secondary structure prediction, in: 2017 3rd IEEE International Conference on Computer and Communications (ICCC), IEEE, 2017, pp. 2713–2717.
- [15] A.W. Senior, R. Evans, J. Jumper, J. Kirkpatrick, L. Sifre, T. Green, C. Qin, A. Židek, A.W.R. Nelson, A. Bridgland, et al., Improved protein structure prediction using potentials from deep learning, *Nature* 577 (2020) 706–710.
- [16] H. Hu, Z. Li, A. Elofsson, S. Xie, A Bi-LSTM based ensemble algorithm for prediction of protein secondary structure, *Appl. Sci.* 9 (2019) 3538.
- [17] J. Jumper, R. Evans, A. Pritzel, T. Green, M. Figurnov, O. Ronneberger, K. Tunyasuvunakool, R. Bates, A. Židek, A. Potapenko, et al., Highly accurate protein structure prediction with AlphaFold, *Nature* 596 (2021) 583–589.
- [18] R.M. Rao, J. Liu, R. Verkuil, J. Meier, J. Canny, P. Abbeel, T. Sercu, A. Rives, Msa transformer, in: International Conference on Machine Learning, PMLR, 2021, pp. 8844–8856.
- [19] L. Wang, T. Jiang, On the complexity of multiple sequence alignment, *J. Comput. Biol.* 1 (1994) 337–348.
- [20] S.F. Altschul, W. Gish, W. Miller, E.W. Myers, D.J. Lipman, Basic local alignment search tool, *J. Mol. Biol.* 215 (1990) 403–410.
- [21] S.R. Eddy, Profile hidden Markov models, *Bioinformatics (Oxford, England)* 14 (1998) 755–763.
- [22] L.S. Johnson, S.R. Eddy, E. Portugaly, Hidden Markov model speed heuristic and iterative HMM search procedure, *BMC Bioinforma.* 11 (2010) 1–8.
- [23] M. Remmert, A. Biegert, A. Hauser, J. Söding, HHblits: lightning-fast iterative protein sequence searching by HMM-HMM alignment, *Nat. Methods* 9 (2012) 173–175.
- [24] S.F. Altschul, T.L. Madden, A.A. Schäffer, J. Zhang, Z. Zhang, W. Miller, D. J. Lipman, Gapped BLAST and PSI-BLAST: a new generation of protein database search programs, *Nucleic Acids Res.* 25 (1997) 3389–3402.
- [25] A.J. Riesselman, J.B. Ingraham, D.S. Marks, Deep generative models of genetic variation capture the effects of mutations, *Nat. Methods* 15 (2018) 816–822.
- [26] A. Riesselman, J.-E. Shin, A. Kollasch, C. McMahon, E. Simon, C. Sander, A. Manglik, A. Kruse, D. Marks, Accelerating protein design using autoregressive generative models, *BioRxiv* (2019) 757252.
- [27] T. Bepler, B. Berger, Learning the protein language: evolution, structure, and function, *Cell Syst.* 12 (2021) 654–669.
- [28] N. Strodthoff, P. Wagner, M. Wenzel, W. Samek, UDSMProt: universal deep sequence models for protein classification, *Bioinformatics* 36 (2020) 2401–2409.
- [29] J. Vig, A. Madani, L.R. Varshney, C. Xiong, R. Socher, N.F. Rajani, BERTology meets biology: interpreting attention in protein language models, in: *ArXiv Preprint ArXiv:2006.15222*, 2020.
- [30] M.J. Zvelebil, J.O. Baum, Understanding Bioinformatics, Garland Science, 2007.
- [31] B. Rost, C. Sander, R. Schneider, PHD-an automatic mail server for protein secondary structure prediction, *Bioinformatics* 10 (1994) 53–60.
- [32] L.J. McGuffin, K. Bryson, D.T. Jones, The PSIPRED protein structure prediction server, *Bioinformatics* 16 (2000) 404–405.
- [33] C. Cole, J.D. Barber, G.J. Barton, The Jpred 3 secondary structure prediction server, *Nucleic Acids Res.* 36 (2008) W197–W201.
- [34] S. Wang, W. Li, S. Liu, J. Xu, RaptorX-Property: a web server for protein structure property prediction, *Nucleic Acids Res.* 44 (2016) W430–W435.
- [35] K. Asai, S. Hayamizu, K. Handa, Prediction of protein secondary structure by the hidden Markov model, *Bioinformatics* 9 (1993) 141–146.
- [36] B. Liu, B. Liu, F. Liu, X. Wang, Protein binding site prediction by combining hidden markov support vector machine and profile-based propensities, *Sci. World J.* 2014 (2014).
- [37] S.C. Potter, A. Luciani, S.R. Eddy, Y. Park, R. Lopez, R.D. Finn, HMMER web server: 2018 update, *Nucleic Acids Res.* 46 (2018) W200–W204.
- [38] X. Cao, W. He, Z. Chen, Y. Li, K. Wang, H. Zhang, L. Wei, L. Cui, R. Su, L. Wei, PSSP-MVIRT: peptide secondary structure prediction based on a multi-view deep learning architecture, *Brief. Bioinform.* 22 (2021) bbab203.
- [39] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [40] M. Mirdita, L. Von Den Driesch, C. Galiez, M.J. Martin, J. Söding, M. Steinegger, Uniclust databases of clustered and deeply annotated protein sequences and alignments, *Nucleic Acids Res.* 45 (2017) D170–D176.
- [41] R. Apweiler, A. Bairoch, C.H. Wu, W.C. Barker, B. Boeckmann, S. Ferro, E. Gasteiger, H. Huang, R. Lopez, M. Magrane, et al., UniProt: the universal protein knowledgebase, *Nucleic Acids Res.* 32 (2004) D115–D119.
- [42] T. Mikolov, K. Chen, G. Corrado, J. Dean, Efficient estimation of word representations in vector space, in: *ArXiv Preprint ArXiv:1301.3781*, 2013.
- [43] Y. Wang, H. Mao, Z. Yi, Protein secondary structure prediction by using deep learning method, *Knowl.-Based Syst.* 118 (2017) 115–123.
- [44] G. Wang, R.L. Dunbrack Jr., PISCES: a protein sequence culling server, *Bioinformatics* 19 (2003) 1589–1591.
- [45] A. Krysztafowicz, A. Barbato, K. Fidelis, B. Monastyrskyy, T. Schwede, A. Tramontano, Assessment of the assessment: evaluation of the model quality estimates in CASP10, *Proteins Struct. Funct. Bioinforma.* 82 (2014) 112–126.
- [46] P. Koehl, M. Levitt, A brighter future for protein structure prediction, *Nat. Struct. Biol.* 6 (1999) 108–111.
- [47] J. Hanson, K. Paliwal, T. Litfin, Y. Yang, Y. Zhou, Accurate prediction of protein contact maps by coupling residual two-dimensional bidirectional long short-term memory with convolutional neural networks, *Bioinformatics* 34 (2018) 4039–4045.

- [48] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimeshein, L. Antiga, et al., Pytorch: an imperative style, high-performance deep learning library, *Adv. Neural Inf. Proces. Syst.* 32 (2019).
- [49] W.L. DeLano, et al., Pymol: an open-source molecular graphics tool, *CCP4 Newsl. Protein Crystallogr.* 40 (2002) 82–92.
- [50] S.F. Ahmed, A.A. Quadeer, M.R. McKay, Preliminary identification of potential vaccine targets for the COVID-19 coronavirus (SARS-CoV-2) based on SARS-CoV immunological studies, *Viruses* 12 (2020) 254.